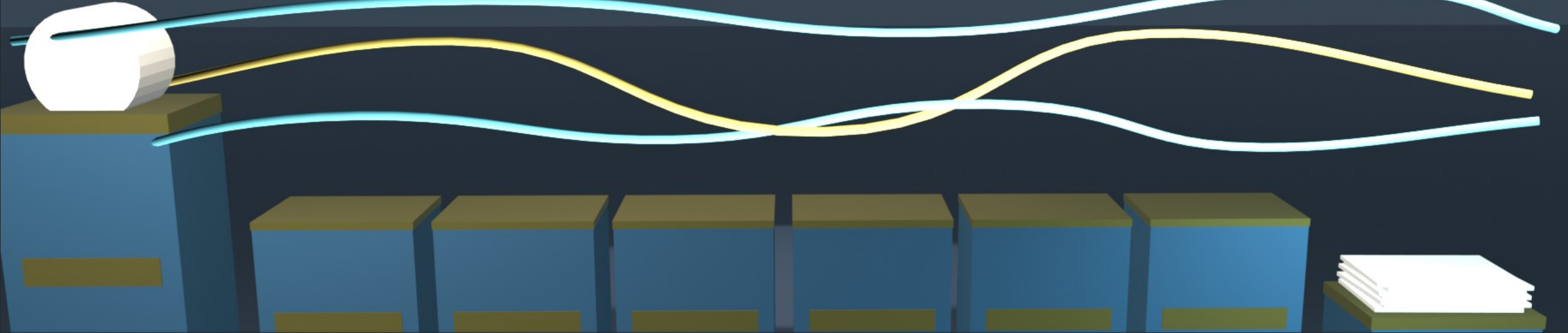




AI-First IT for a Growing Manufacturer

My 90-Day Vision — Enmanuel D. Mejia

Mid-size commercial print manufacturer · October 2026



Created by Interstitium Labs

Print + packaging manufacturing meets AI

Predictive maintenance — press telemetry can flag bearing, roller, and blanket wear before it costs a run.

Prepress automation — the job data already flowing from MIS to prepress can drive preflight, imposition checks, and plate release with fewer touches.

Vision-based print inspection — a camera plus a model at the delivery end catches color and register drift in-line, before waste piles up.

Standardized job data — clean, normalized data across the MIS and the floor is what makes every later model trustworthy. Garbage in, garbage out.

Responsible-AI guardrails — no client artwork in public tools, segmented OT, and a human in the loop on every decision that ships.

The win: measurable operational impact — not experiments.

From the company's public newsroom

THE COMPANY

- Commercial print & packaging manufacturer — Orlando, FL; founded 2007
- 100–150 employees; three adjacent production buildings
- Florida's Best Printer 2026 (Florida Print Awards)
- FPA 2026 Silver, Shelf Impact — pet-nutrition brand pouch
- CEO an AI Board Champion in YPO's Manufacturing Excellence Network

THE 2026 GROWTH WAVE — FROM YOUR NEWSROOM

- Heidelberg Versafire LV + third POLAR cutter — announced April 2026
- HP Indigo 200K digital press — announced April 2026
- Sun Centre XL-DR pouching machine — announced May 2026
- Hudson-Sharp Ares 400-SUP, custom dual registration — ~June 2026
- Third building: 15,000 sq ft, folding cartons + inline carton camera — announced January 2026
- Heidelberg Speedmaster CD 102-6+L (6-color 40", inline coater) — the worked example in this deck

THE THESIS — The company is already AI-forward for a mid-size printer. My job: take that program from strong pilots to a governed, company-wide architecture.

Your public AI pillars — Heidelberg Color Assistant Pro, HP onboard AI/ML, Signa Station, Prinect PAT (13,000+ connected presses), Enfocus Switch, Aleyant Pressero with AI-driven job routing — plus the internal program connecting core systems (Jay Kahn, LinkedIn, Aug 2026).

Interview question for Friday: I'd love to confirm the 2026 additions on the production floor.

Five pillars, one lifecycle

1 · AI-DRIVEN WORKFLOW & AUTOMATION

The headline. Hunt repetitive work and automate it with AI that proves its value.

2 · TECHNICAL SUPPORT & SYSTEM ADMINISTRATION

Mixed Windows/Mac/Linux/Windows Server, AD/LDAP, patching, backups — PowerShell + Bash fluent.

3 · INFRASTRUCTURE & NETWORKING

Hardwired production-floor connections plus office systems — segmented and reliable.

4 · DOCUMENTATION & PROCESS IMPROVEMENT

Ticketing discipline like JIRA; docs that make improvements stick and transfer.

5 · END-USER TRAINING

Coaching the team on responsible, productive AI use.

THE LIFECYCLE I RUN ON EVERY PILLAR

discover & assess → pilot → roll out → implement → train → maintain & continuously improve

Two tiers — the right model for each job

MODEL ROUTER / GATEWAY

Inspects every request — data sensitivity, task type, cost, latency — and sends it to the right tier.

One chokepoint for policy, logging, and cost control.

Fail closed: sensitive data can only resolve to Tier 1 — default-deny to the building.

TIER 1 — OPEN-WEIGHTS, SELF-HOSTED ON-PREM

- Customer artwork & packaging files
- Job tickets & Tharstern MIS data
- Color profiles & Image Control data

Fortune 500 packaging work can never leak into public tools.

TIER 2 — FRONTIER MODELS VIA API

- Heavy reasoning & troubleshooting
- Code generation for automation

Only data that is safe to send — never client artwork.

TRADE-OFFS, MADE EXPLICIT

PRIVACY — Tier 1 keeps everything in-house; Tier 2 sends data to a vendor.

LATENCY — Tier 1 infers locally with no egress; Tier 2 adds a network round-trip.

COST — Tier 1 is a fixed GPU cost with unlimited use; Tier 2 is metered per token.

CAPABILITY — Tier 2 brings frontier reasoning; Tier 1 is tuned to our own workflows.

Jev — LLMs generate, Jev decides, software governs

WHAT JEV IS

- TypeSafe's "System One" decision model — launched September 2026
- Not a generative LLM: answers typed questions (Choice / Score / Noul)
- Returns structured values with calibrated probabilities
- Via API: POST `api.typesafe.ai/v1/systemone`, route "jev-latest"
- Does not draft text, write code, or plan — LLMs still do that

WHERE IT FITS

- Tier 1 on-prem: private data + generation
- Tier 2 frontier APIs: heavy reasoning + coding
- Jev between them: the structured decisions
- Extends your pillars: Switch/Pressero route, PAT optimizes — Jev judges
- "LLMs generate, Jev decides, software governs."

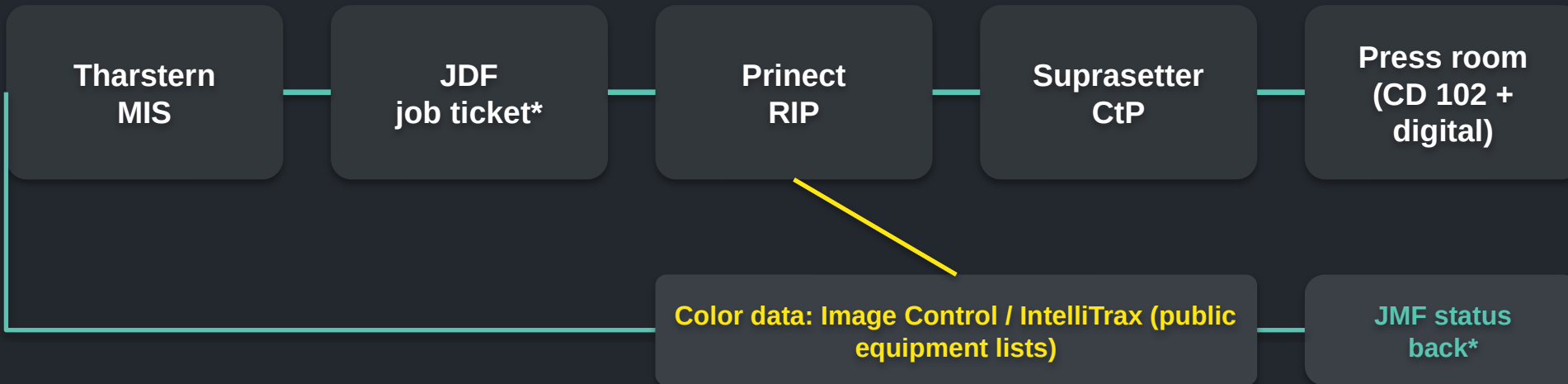
EXAMPLE FLOW

1. A job ticket arrives (JDF/JMF in a typical setup — to confirm)
2. Jev scores its priority and route (calibrated probability)
3. The JDF/MIS agent acts on the scored decision
4. A human approves — every time

HONEST FRAMING — A PILOT-PHASE RECOMMENDATION

- The recommendation: Jev as the decision layer — one 30-day pilot on order triage (50–300 labeled items) before anything touches production routing.
- Access is waitlist / early-access; pricing announced at \$0.042 per million input tokens — the pilot starts when access and written data terms land.
- TypeSafe's no-training-on-customer-data / zero-data-retention claims are UNVERIFIED on primary sources — artwork stays on-prem regardless.
- Contingency, not an open option: an open-weights classifier stands behind the pilot, so the program never stalls on a waitlist.

The backbone: sysadmin + infrastructure



* Inferred — typical Prinect/Tharstern shop setup; I confirm in Phase 0.

WHAT I'D ASSESS IN PHASE 0 — THEN OWN

- Phase 0 assessment: domain/identity, VLANs across the three buildings, backup/DR, monitoring — unverified publicly; I measure, I don't guess
- Segment production/OT from office IT — dedicated production VLANs; press consoles off the office LAN (proposed, Phase 1)
- Terabyte-scale artwork storage with versioning; tested 3-2-1 backups of RIP queues, the MIS database, calibration data (proposed)
- Patch windows that never stop a press run; vendor-pinned console builds hardened per Heidelberg guidance — never freelanced
- Vendor remote-service paths (e.g., Heidelberg eCall / Remote Service — to confirm): time-boxed, on-request, logged

What keeps AI safe, measurable, and reversible

Local inference serving — open-weights models served on company hardware — versioned, monitored, and restartable.

Prompt & version management — every prompt, model, and config pinned and reviewed like code — no silent changes.

Guardrails & content filters — what a model may see, say, and touch is declared up front — not hoped for.

Audit logging of every AI action — who asked, what the model saw, what it did — a full trail on every request.

Cost tracking per use case — tokens, GPU-hours, and wall-clock time attributed to each pilot and agent.

Eval harness — every automation regression-tested before it touches production — failures block the rollout.

If it isn't logged, it didn't happen; if it can't be rolled back, it doesn't ship.

THE AGENTS

Task-specific agents, grounded on your floor

PREPRESS PREFLIGHT AGENT

Floor: prepress

IN — JDF ticket + incoming PDF artwork

TOUCHES — PitStop rules, Switch flows, plate queue

GATE — Prepress operator confirms every exception release

BACK — Re-release the previous version from the versioned queue

JDF / MIS MONITORING AGENT

Floor: Prinect ↔ Tharstern bridge

IN — Ticket stream (JDF/JMF — typical setup, to confirm)

TOUCHES — Read-only MIS queries, alerts, ticket notes

GATE — Supervisor confirms before any ticket is corrected

BACK — Full ticket history; revert to last-known-good

TROUBLESHOOTING CO-PILOT

Floor: IT admin

IN — Server, RIP, and console logs + monitoring alerts

TOUCHES — Read-only log search, runbook library

GATE — The admin approves every action — the agent never runs one

BACK — No autonomous changes, so nothing to roll back; all suggestions logged

DOCUMENTATION AGENT

Floor: everywhere

IN — Ticket-system tickets + resolutions (platform to confirm)

TOUCHES — Ticketing API, runbook library

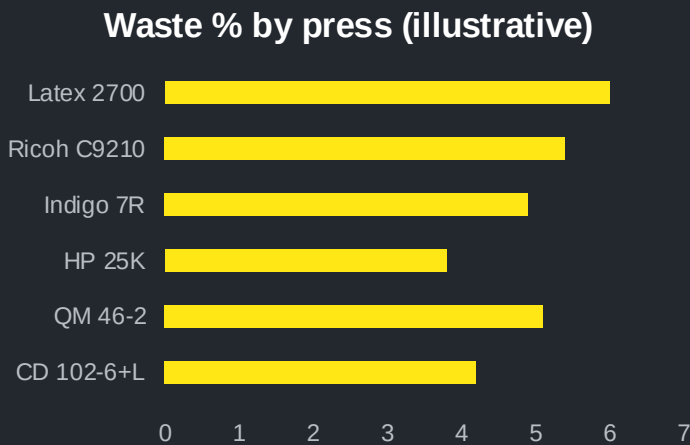
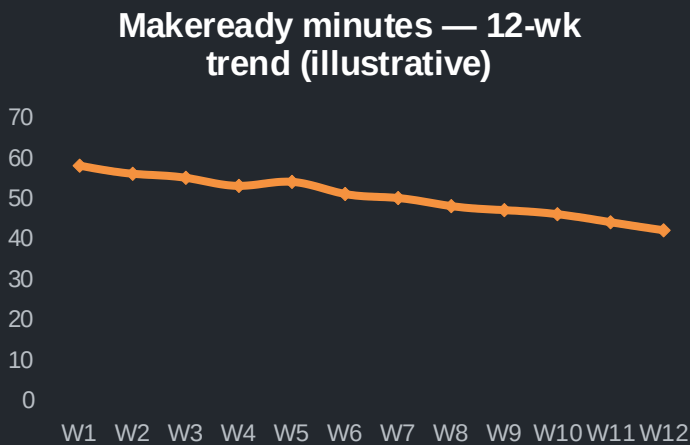
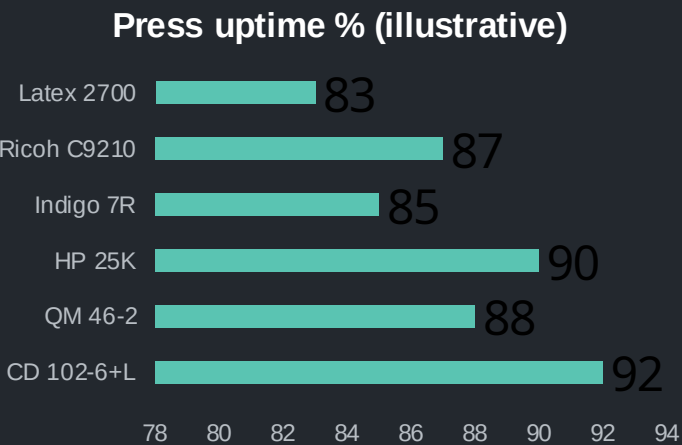
GATE — Team lead reviews every runbook before publish

BACK — Versioned docs; restore the previous version in one click

Every agent: a human in the loop on anything that ships — and a way back.

What the floor could see every morning

ILLUSTRATIVE TARGETS — NOT THE COMPANY DATA



AI PROJECT PIPELINE

- PILOT
- IN PROGRESS
- PLANNED

Closed-loop color analytics — drift alerts from Image Control data

OT VLAN + telemetry capture across all presses

Preflight automation · predictive maintenance model · MIS anomaly detection

Each KPI ties to a measurable outcome — time saved, waste avoided, makeready shortened.

Phases 0–2: audit, foundation, pilots

PHASE 0 — AUDIT

Weeks 1–4

OWNER: Me — I own the audit end to end

DEPENDENCIES: Floor-walk access, MIS & prepress data export permissions, time with prepress, press, and IT staff

RISKS: Baselines skew if the floor is too busy — mitigate by shadowing shifts, not interviews alone

SUCCESS: Catalog of every repetitive task with hour baselines, ranked by effort vs. impact; first pilots chosen

PHASE 1 — FOUNDATION

Inference, router, OT segmentation

OWNER: Me — architecture, build, and change control

DEPENDENCIES: GPU hardware procured; OT change windows that never stop a press run; vendor remote-access policy aligned

RISKS: Legacy consoles resist segmentation — mitigate with staged cutover and a rollback plan

SUCCESS: Router + audit logging live on 100% of AI requests; presses on segmented OT with zero production downtime

PHASE 2 — PILOTS

2–3 agents live, measured before/after

OWNER: Me — pilot delivery and measurement

DEPENDENCIES: Baseline data from Phase 0; eval harness from Phase 1; operator buy-in; Jev early-access (waitlist)

RISKS: Pilot scope creep — mitigate with a fixed 4-week measurement window; open-weights contingency built in parallel if Jev access stalls

SUCCESS: Measured before/after per pilot: time saved, waste avoided; documented go / no-go decisions

Phases 3–4: rollout, then never stop improving

PHASE 3 — ROLLOUT

Fleet-wide: CD 102, Indigo 200K, pouch line, prepress

OWNER: Me — rollout sequencing and eval gates

DEPENDENCIES: Measured pilot results; operator training complete; eval suite per press

RISKS: Model behavior varies per press — mitigate with per-press tuning and eval gates

SUCCESS: Every press and line on the KPI dashboard; each rollout passes its eval gate

PHASE 4 — CONTINUOUS IMPROVEMENT

Evals, refresh cadence, responsible-AI training

OWNER: Me — standing program ownership

DEPENDENCIES: GPU maintenance budget; quarterly training slots with the team

RISKS: Model drift over time; staff turnover — mitigate with the eval harness and documented runbooks

SUCCESS: Quarterly model refresh with eval pass-rate held; team trained on responsible AI use

RISKS & HONEST TRADE-OFFS

Every risk named — every mitigation owned.

OT SAFETY — An agent never actuates a press without a human gate — suggestions, not control.

DATA LEAKAGE — The on-prem tier is the answer — sensitive data never leaves the building.

VENDOR LOCK-IN — Open weights plus an API-agnostic router keep either tier swappable.

MODEL DRIFT — The eval harness catches it; the refresh cadence fixes it.

Competitive Benchmark

Ten entries graded on evidence — then the one system I would build

COMPETITOR	WHAT THEY DEPLOY	EVIDENCE	HOW THE FABRIC MATCHES / EXCEEDS
Heidelberg	Intellistart 3 (<30s, vendor claim) · Color Assistant Pro · Portal NL analytics (3,000+ shops) · Prinect Touch Free beta — not an MIS	Strong	Keep Heidelberg's machine layer; build the cross-vendor learning layer it leaves untouched
HP PrintOS / Indigo	AAA 2.0 ML defect detection (Elanders, vendor-reported) · Nio chatbot · AI Order Intake · 400+ Indigo inspection installs	Strong	Strongest press-floor AI — but Indigo-locked; the fabric treats Indigo as one device class
Konica Minolta IQ-501	Inline closed-loop QC accessory (vendor claim: profiling 30 → 5 min); FastPrint: halved turnarounds	Strong	Extend inline QC across ALL presses — an accessory is not a workflow brain
Enfocus PitStop/Switch	Ubiquitous deterministic preflight; “AI-powered” is show-floor language — no neural preflight ships	Strong	Keep as the guardrail base; add ML above it, learning from shop prepress history
BMW AIQX	1,000+ camera/sensor units at Landshut; live at three more plants	Strong	The template for the edge-vision layer — the same pattern, smaller and cheaper
GE Appliances	800+ agents on Gemini Enterprise (Sept 2026); one cut back orders 25% (company-reported)	Strong	Same agentic pattern, scoped agents with human gates — sized for a three-building printer
Goldman Sachs	46,500+ staff assistant routing across OpenAI / Gemini / Llama by task (CNBC, Jan 2025)	Mod-Strong	The exact precedent for the task-typed model gateway
Meta Llama Guard	Llama Guard 4 + Prompt Guard 2 back the production Moderations API; NeMo Guardrails analog	Strong	“Small model decides” is production-standard — the Jev-style layer is the next step
ANZ / Omada / France	Self-hosted Llama/Mistral for compliance & sovereignty (10,000 agents, 8 ministries)	Moderate	The same sovereignty logic: artwork, tickets, mailing lists never leave the building
Jev (TypeSafe)	Typed decisions (Choice/Score/Noul); launched Sept 15, 2026; ~\$0.0004/decision; ~110–175 ms	Weak	Evaluated plug-in, not dependency — Llama Guard/NeMo fallback runs today
SOVEREIGN HYBRID AI FABRIC — on-prem open weights · typed gateway · decision layer · edge vision · OT-segmented · Tharstern system of record			

Full detail: AI-Vision-UltraPlan (companion document).

My own year-one scorecard — personal commitments

Automate the repetitive work — manual tasks identified and automated, with time saved tracked and reported.

Embed AI in the daily craft — AI tools used in real troubleshooting, documentation, and monitoring — not demoed.

Keep the environment smooth — a mixed environment that runs clean, with issues resolved fast and independently.

Train confident users — the team comfortable using AI responsibly and productively in their own work.

Document everything — clear, current documentation for procedures, AI workflows, and every system change.

Guardrails before models

Vendor remote access — time-boxed, on-request only, and logged. Remote-service paths open when asked and close when done — never standing access.

OT segmentation — press consoles on their own VLANs with no direct internet path. AI inference near the press stays local or on the segmented network.

Client artwork stays private — no job files, artwork, or customer data in public AI tools. Ever. Evaluation uses synthetic or fully-consented data.

Tested patching on legacy consoles — staged, verified updates on a schedule that never interrupts a press run.

Monitoring and alerting discipline — every AI system watched in production; alerts tuned so real signals get seen, and a human signs off on anything that ships.

Verified background — nothing invented

5+ years progressive hands-on IT experience — including Data Vimenca, IT Operations Engineer II / IT Support Engineer II (contract, remote), Aug 2021 – Apr 2026.

CompTIA A+ CE — certified, valid through April 30, 2027.

Mixed-OS fluency — Windows / Mac / Linux / Windows Server; PowerShell + Bash automation and scripting.

Hands-on builder — I design, build, and operate production-grade infrastructure in my own independent lab — the discipline behind this deck's architecture.

Full work history and references available in my résumé packet.



Thank you — I want to earn this.

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Looking forward to seeing the floor on Friday.